



MRSx241-DEV-V1

mmWave Radar Development Board

User Guide

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1 Overview

The RSx241-DEV-BOX development kit from Possumic Technology is an application development platform integrating an ultra-low-power 60 GHz FMCW mmWave radar sensor chip. It facilitates solution evaluation and design. The kit includes MRS6241-series modules (MRS6241-P2828-M62 and MRS6241-P2840-M81), MRS7241-series modules (MRS7241-P2828-M62 and MRS7241-P2840-M81), and an MRSx241-DEV-V1 development board. It allows developers to connect a wide range of peripherals for diverse application development, especially in smart industry and intelligent driving. Its multiple-transmit, multiple-receive antenna configuration delivers excellent detection performance for product applications.

This document describes how to use the MRSx241-DEV-V1 development board, including component, power-supply, and inter face information.

2 Hardware Introduction

2.1 Hardware Block Diagram

The main components and application connections of the MRSx241-DEV-V1 are shown in the following figure:

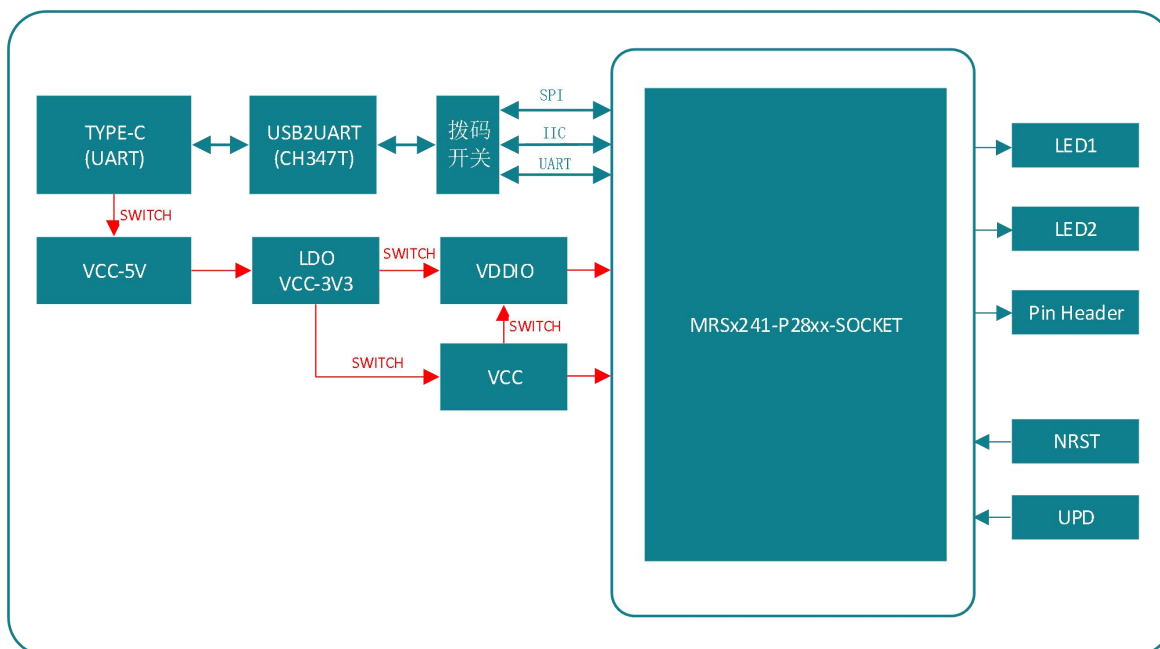


Figure 2-1 MRSx241-DEV-V1 Hardware Connection Block Diagram

2.2 Component Introduction

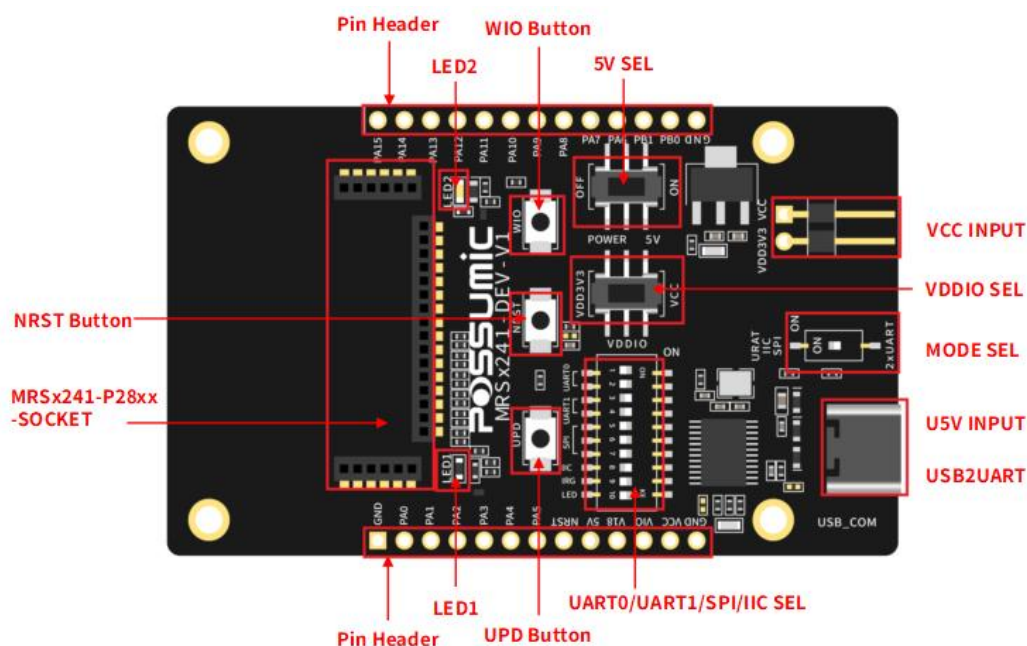


Figure 2-2 Development Board Components

Table 2-1 Development Board Component Descriptions

Component		Description
Module Interface	MRSx241-P28xx-SOCKET	Connects an MRS6241-P2828-M62/MRS6241-P2840-M81 mmWave radar module/ MRS7241-P2828-M62/MRS7241-P2840-M81
Buttons	UPD Button	Upgrade/Download button With the board powered off, press and hold the UPD Button, and then connect the USB cable to power on the board. The chip starts in firmware download mode; download the firmware through UART0.
	NRST Button	Module reset button
	WIO Button	Module sleep/wake trigger button
Power Supply	U5V INPUT	Provides 5 V from Type-C VBUS and also serves as the UART-to-USB connection to a PC.
	5V SEL	Development-board main VCC-5V power switch: ➤ Set to ON (default): Turns on the VCC-5V supply. ➤ Set to OFF: Turns off the VCC-5V supply.
	VDDIO SEL	Selects the source for module VDDIO, which supplies the module I/Os and peripherals: ➤ Set to VCC (default): Uses VCC as the supply. ➤ Set to VDD3V3: Uses VDD3V3 as the supply.
	VCC INPUT	Module VCC power switch; VCC is the main radar-module supply: ➤ Install the jumper cap (default): Uses VDD3V3 from the on-board LDO, whose input is VCC-5V. ➤ Remove the jumper cap: To use an external DC power source, remove the jumper cap and connect the external supply to the VCC pin on the Pin Header.
Other	USB2UART	Type-C USB interface used for development-board power and UART/IIC/SPI communication. When MODE SEL is set to OFF, the USB interface supports dual UARTs. After connection to a PC, the devices are identified as USB Serial Device (COM*), where * is the COM-port number. The higher number corresponds to radar UART0, and the lower number corresponds to radar UART1. When MODE SEL is set to ON, the USB interface supports one UART, IIC, and SPI. After connection to a PC, the device is identified as USB Serial Device (COM*).
	LED	Application indicator LEDs. Powered by VDD3V3. LED1 is controlled by PA6, and LED2 is controlled by PA11.
	MODE SEL	CH347T operating-mode selection (DIP switch): ➤ ON (default): USB to one high-speed UART + USB to 2-wire IIC + USB to 4-wire SPI ➤ OFF: USB to two high-speed UARTs Note: The ON direction is indicated on the DIP switch.
	UART0/UART1/SPI/IIC SEL	MRS6240 communication-mode selection (DIP switch): ➤ Set 1 and 2 to ON and the others to OFF: Connects UART0 to Type-C ➤ Set 3 and 4 to ON and the others to OFF: Connects UART1 to Type-C

		➤ Set 5, 6, and 7 to ON and the others to OFF: Connects SPI to Type-C ➤ Set 8 to ON and the others to OFF: Connects IIC to Type-C ➤ Set 9 to ON and 10 to OFF: PA6 reports interrupt data to the USB chip ➤ Set 10 to ON and 9 to OFF: PA6 is used to drive LED1
	Pin Header	All available module I/Os (except the dedicated Flash SPI interface) and power rails (VCC, VDDIO, and V13) are routed to this pin header.

2.3 Power-Supply Options

1. Development-board power:

1) USB_COM interface power: Set the VCC-5V DIP switch to ON (default).

2. The module VCC supply has two options:

1) Development-board VCC-3.3 V supply: Install the VCC INPUT jumper to connect VDD3V3 and VCC (default).

2) External supply: With VCC INPUT disconnected, apply power to the VCC pin on the Pin Header.

3. The development-board and module VDDIO supply has two options:

1) Development-board VCC-3.3 V supply: Set the VDDIO DIP switch to VDD-3V3 (default).

2) VCC supply: Set the VDDIO DIP switch to VCC.

2.4 Pin Header

The names and functions of the pin headers on both sides of the development board are listed below:

Table 2-2 Pin Header Signal Definitions

No.	Name	Type	Function
1	GND	G	Development-board ground
2	PA0	IO/WUP	PA0,WIO0,SPI0_CS,UART1_TX,CANFD_TX,PWM0
3	PA1	IO/WUP	PA1,WIO1,SPI0_CLK,UART1_RX,CANFD_RX,PWM1
4	PA2	IO/WUP	PA2,WIO2,SPI0_MOSI,UART1_RTS,I2C0_SDA,UART2_TX/LIN_TX,PWM2
5	PA3	IO/WUP	PA3,WIO3,SPI0_MISO,UART1_CTS,I2C0_SCL,UART2_RX/LIN_RX,PWM3
6	PA4	IO/WUP	PA4,WIO4,SPI0_HOLD,I2C0_SDA,PWM4
7	PA5	IO/WUP	PA5,WIO5,SPI0_WP,I2C0_SCL,PWM5
8	NRST	I	Radar-module reset signal
9	5V	PI	Development-board 5 V supply; USB power is used by default
10	V18	PI/PO	Module V13 supply; this rail measures 1.5 V when the internal LDO is used
11	VDDIO	PI	Development-board VDDIO supply
12	VCC	PI	Development-board VCC supply
13	GND	G	Development-board ground

14	GND	G	Development-board ground
15	PB0	IO/WUP	PB0,WIO16,UART0_TX,PWM0
16	PB1	IO/WUP	PB1,WIO17,UART0_RX,PWM1
17	PA6	IO/WUP	PA6,WIO6,UART1_RTS,UART2_TX,I2C1_SDA,PWM6,GPADC6
18	PA7	IO/WUP	PA7,WIO7,UART1_CTS,UART2_RX,I2C1_SCL,PWM7,GPADC7
19	PA8	IO/WUP	PA8,WIO8,UART1_TX,I2C1_SCL,PWM0,GPADC0
20	PA9	IO/WUP	PA9,WIO9,UART1_RX,I2C1_SDA,PWM1,GPADC1
21	PA10	IO/WUP	PA10,WIO10,I2C1_SDA,CANFD_TX,PWM2,GPADC2
22	PA11	IO/WUP	PA11,WIO11,I2C1_SCL,CANFD_RX,PWM3,GPADC3
23	PA12	IO/WUP	PA12,WIO12,UART2_RX,UART1_RTS,PWM4,GPADC4
24	PA13	IO/WUP	PA13,WIO13,UART2_TX,PWM5,GPADC5
25	PA14	IO/WUP	PA14,WIO14,I2C0_SDA,UART2_RX,CANFD_TX,PWM6
26	PA15	IO/WUP	PA15,WIO15,I2C0_SCL,UART2_TX,CANFD_RX,PWM7

Note:

- **G** denotes ground.
- **PI** denotes a power input.
- **PO** denotes a power output.
- **GPIO** denotes a general-purpose digital pin.
- **I** denotes an input pin.
- **WUP** denotes a digital pin that can retain its state or wake the system in standby mode. WIOx denotes the sequential number of a WUP-type I/O.

3 Hardware Operation

3.1 Required Hardware

Prepare the following hardware to use the MRSx241-DEV-V1 development board:

- MRS6241-P2828-M62, MRS6241-P2840-M81, MRS7241-P2828-M62, or MRS7241-P2840-M81 radar module
- Type-C USB data cable
- Computer with the CH347T UART driver installed

3.2 Module Connection

The MRS6241-P2840-M81 radar module is used as an example. The connection is shown below.

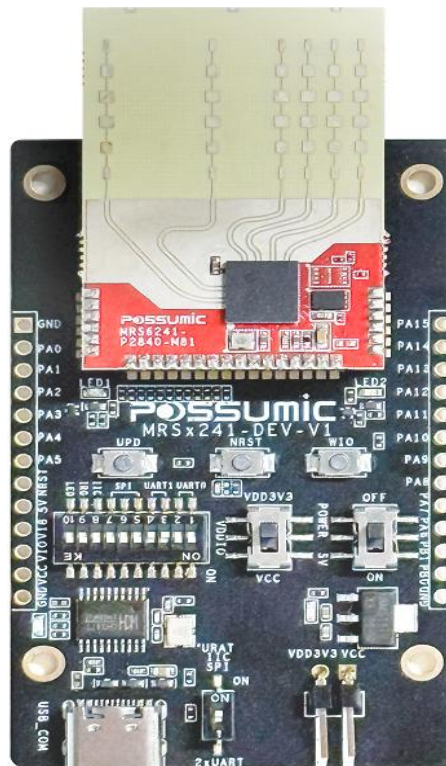


Figure 3-1 MRS6241-P2840-M81

3.3 Basic Operation

- **Step 1:** Configure the development-board power using the default connections described under [Power Supply in Table 2-1, Development Board Component Descriptions](#).
- **Step 2:** Configure the signal path using the default **MODE SEL** and **UART0/UART1/SPI/IIC SEL** settings described in [Table 2-1, Development Board Component Descriptions](#).
- **Step 3:** Insert the MRS6241-P2840-M81 radar module.
- **Step 4:** Connect the PC to the U5V TYPE-C port on the development board with a Type-C data cable. Two serial devices appear on the PC. The higher COM-port number corresponds to radar UART0, and the lower number corresponds to radar UART1.
- **Step 5:** Operate the board with the PC host tool.

3.4 Power-Consumption Measurement

- **Step 1:** Set 5V SEL to ON and VDDIO SEL to VCC, and remove the VCC INPUT jumper cap.
- **Step 2:** Power the U5V TYPE-C port with a Type-C data cable.
- **Step 3:** Use an external high-precision power measurement instrument to supply VCC and GND on the Pin Header, and record the current.

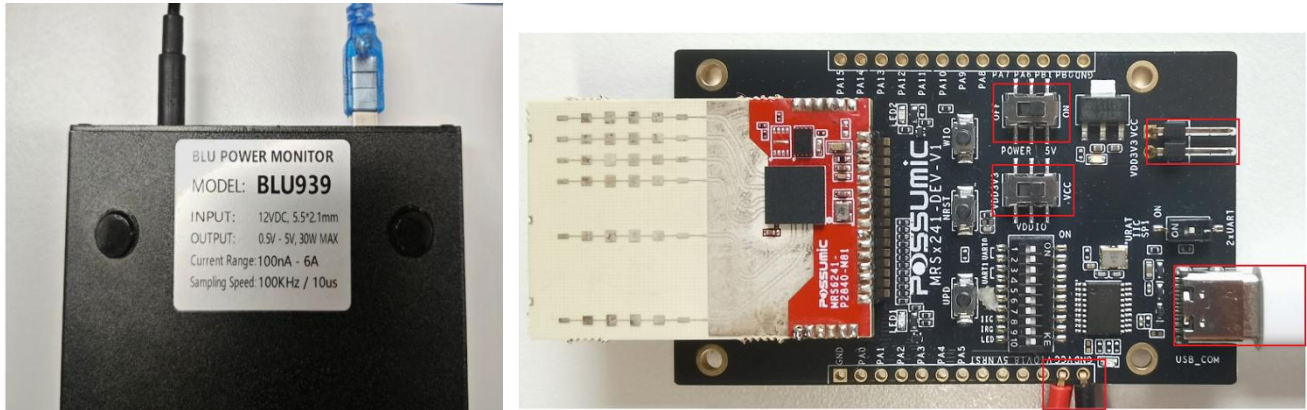


Figure 3-5 Power Measurement Instrument and Hardware Setup

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